

# Lab Session 11

## *Agile Sprint Planning Using Trello*

### 1. Introduction

In modern software development, Agile methodology is widely used to manage projects that require flexibility, rapid delivery, and collaboration. Unlike traditional waterfall models, Agile emphasizes iterative development, frequent feedback, and adaptive planning.

The Scrum framework is one of the most popular Agile frameworks. It organizes work into time-boxed iterations called sprints, where the team delivers working software incrementally.

The goal of this lab is to teach students how to plan and manage a sprint using Scrum principles, visualize tasks using Trello, and monitor progress using priority and burn-down charts.

### 2. Learning Objectives

By completing this lab, students will learn to:

1. Understand **Agile methodology and Scrum framework**
2. Define **Sprint Goals** aligned with project objectives
3. Select **user stories** from a product backlog and prioritize them
4. Break user stories into actionable tasks
5. Assign tasks to team members effectively
6. Estimate effort using hours or story points
7. Set up and manage a Trello Sprint Board
8. Draw priority charts and burn-down charts
9. Conduct a reflection on sprint planning, task management, and coordination

### 3. Agile Methodology

Agile methodology is a **set of principles for software development** that prioritize:

- **Individuals and interactions** over rigid processes and tools
- **Working software** over extensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a strict plan

#### Key Benefits of Agile:

- Faster delivery of functional software
- Improved customer satisfaction through frequent feedback
- Adaptable to changing requirements
- Increased team collaboration and accountability

## Agile Practices:

1. **Iterative Development:** Break the project into small increments delivered in sprints.
2. **Continuous Feedback:** Engage stakeholders regularly to validate functionality.
3. **Adaptive Planning:** Reassess priorities after each iteration.
4. **Collaborative Teams:** Cross-functional teams work together on design, development, and testing.

## 4. Scrum Framework

Scrum is a lightweight framework for managing complex projects using Agile principles.

### 4.1 Scrum Roles

| Role             | Responsibilities                                                                                         |
|------------------|----------------------------------------------------------------------------------------------------------|
| Product Owner    | Manages the product backlog, prioritizes stories, ensures team builds the right features                 |
| Scrum Master     | Facilitates Scrum processes, removes obstacles, ensures team follows Agile practices                     |
| Development Team | Implements and tests tasks, self-organizing, responsible for delivering potentially shippable increments |

### 4.2 Scrum Artifacts

1. **Product Backlog** – A complete list of features, enhancements, and bug fixes needed for the product.
2. **Sprint Backlog** – Subset of the product backlog selected for a sprint.
3. **Increment** – Completed, functional product delivered at the end of a sprint.

### 4.3 Scrum Ceremonies

1. **Sprint Planning:** Define sprint goal, select stories, break them into tasks, assign work.
2. **Daily Stand-up:** Quick daily meeting to review progress and obstacles.
3. **Sprint Review:** Demonstrate the completed increment to stakeholders.
4. **Sprint Retrospective:** Reflect on the sprint to improve team processes.

## 5. Sprint Planning

Sprint Planning ensures that the development team knows:

- **What to build** (Sprint Goal)
- **How to build it** (Tasks, assignments, and dependencies)
- **How long it will take** (Effort estimation)

## 5.1 Steps in Sprint Planning

1. **Define Sprint Goal** – Clear objective for the sprint, aligned with product vision.
2. **Select User Stories** – Choose items from the product backlog achievable within the sprint.
3. **Prioritize Stories** – Identify high-value stories first.
4. **Break Stories into Tasks** – Smaller tasks are easier to estimate and assign.
5. **Estimate Effort** – Use hours or story points for planning capacity.
6. **Assign Tasks** – Allocate work based on team members' expertise.
7. **Plan Sprint Board** – Visualize tasks in **To Do, In Progress, Done** columns.
8. **Visualize Progress** – Priority charts and burn-down charts track performance.

## 6. Product Backlog

Before we can begin a sprint, we need to **prepare a product backlog**.

### What is a Product Backlog?

- A product backlog is a **living document** listing all features, enhancements, bug fixes, and tasks that could be done for a project.
- Each item is written as a **user story**, e.g., “As a user, I want to log in so that I can access my account.”
- Stories should be **prioritized** by value, risk, and dependencies.

### Breaking Stories into Tasks:

- Each story should be broken into **smaller, manageable tasks**.
- Tasks are easier to **estimate, assign, and track** than full user stories.
- Example tasks for a “Login” story:
  - o Design login form UI
  - o Implement backend authentication
  - o Connect UI with backend
  - o Test login functionality

### Example Product Backlog Table

Students create the **Product Backlog Table** in Word.

| ID  | User Story         | Description                   | Priority |
|-----|--------------------|-------------------------------|----------|
| US1 | User Login         | Implement login functionality | High     |
| US2 | User Registration  | Sign-up form & validation     | High     |
| US3 | Profile Management | Edit profile details          | Medium   |
| US4 | Dashboard          | Display recent activity       | Medium   |
| US5 | Notifications      | Alerts for updates            | Low      |

## 7. Breaking Stories into Tasks

- Breaking stories into tasks ensures clarity, accurate estimation, and easier tracking.
- Each task should ideally be **completable within 1–2 days**.

**Example: US1 – User Login**

| Task ID | Task Description                 | Assigned To | Estimated Hours | Status |
|---------|----------------------------------|-------------|-----------------|--------|
| T1.1    | Design login form UI             | Front-end   | 3               | To Do  |
| T1.2    | Implement backend authentication | Back-end    | 4               | To Do  |
| T1.3    | Connect UI with backend          | Front-end   | 2               | To Do  |
| T1.4    | Test login functionality         | QA          | 2               | To Do  |

## 8. Priority Chart

- Priority charts **visualize which stories/tasks are most important**.
- Helps the team focus on **high-value features first**.

Students count tasks according to priority.

### Steps

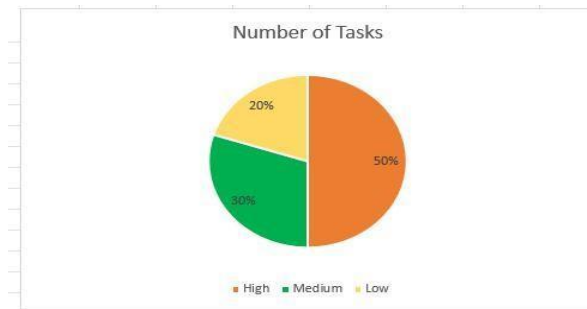
1. Open **Microsoft Excel**
2. Enter the table
3. Select the table
4. Click **Insert**
5. Choose **Bar Chart or Pie Chart**

Format:

X-axis → Priority

Y-axis → Number of Tasks

| Priority | Number of Tasks |
|----------|-----------------|
| High     | 5               |
| Medium   | 3               |
| Low      | 2               |



## 9. Trello Sprint Board

- Trello is a **Kanban-style board** to manage tasks.
- Columns (lists) represent **workflow stages**.

Students manage task using trello.

### Step-by-step Instructions:

1. Create a new Trello board.
2. Create **columns**: To Do | In Progress | Done
3. Add **cards** for each task.
4. Assign **team members** to each card.
5. Move cards from **To Do** → **In Progress** → **Done** as work progresses.

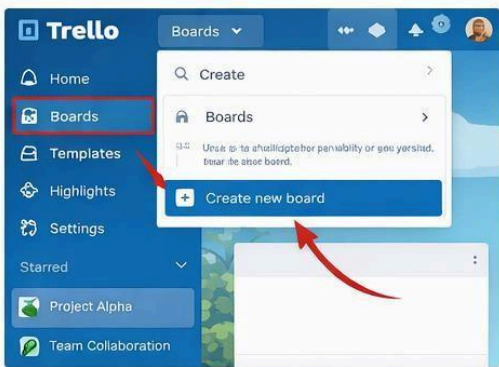
### Example Layout:

- **To Do:** T1.1, T1.2, T2.1
- **In Progress:** T1.3
- **Done:** T1.4

**Tip:** Use **labels** to show priority, story type, or department.

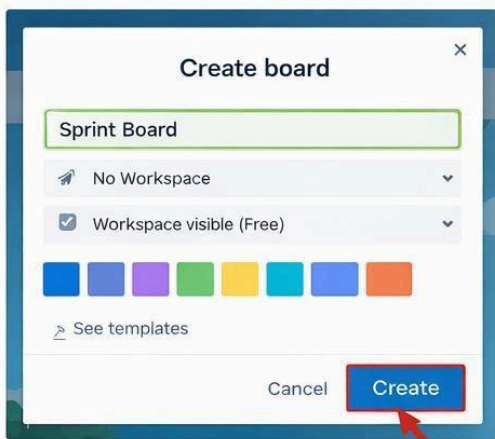
### 1 Create a New Board

- Click the “**Boards**” option in Trello’s sidebar
- Select “**Create new board**” from the dropdown



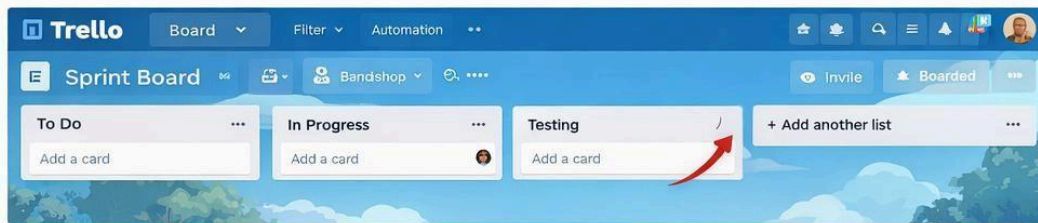
### 2 Name Your Board

- Enter your board’s name, like “**Sprint Board**”
- Click “**Create**” to make your new board.



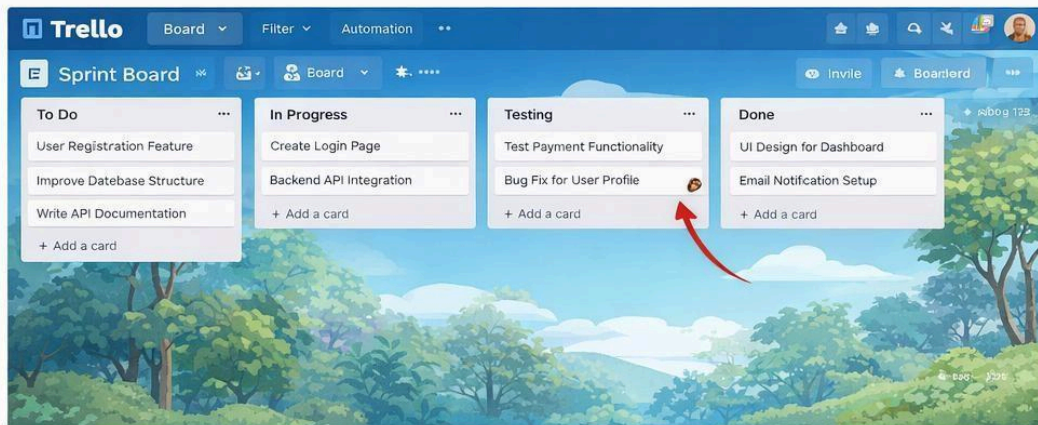
### 3 Add Lists to the Board

- Create lists like “To Do”, “In Progress,” “Testing,” and “Done”
- Click “Add a list” button to add each list.



### 4 Add Tasks and Start Sprint

- Add tasks to the “To Do” list for your sprint.
- Move tasks through the lists as they progress.



## 10. Burn-down Chart

- Burn-down charts track **remaining work vs sprint days**.
- Helps monitor progress and identify if the team is on schedule.

Students track **remaining work each day**.

#### Step-by-step Instructions:

1. Record **estimated hours** for each task in Trello (custom field or card description).
2. At the end of each day, calculate **remaining hours**.
3. Create a table in Excel or Sheets:

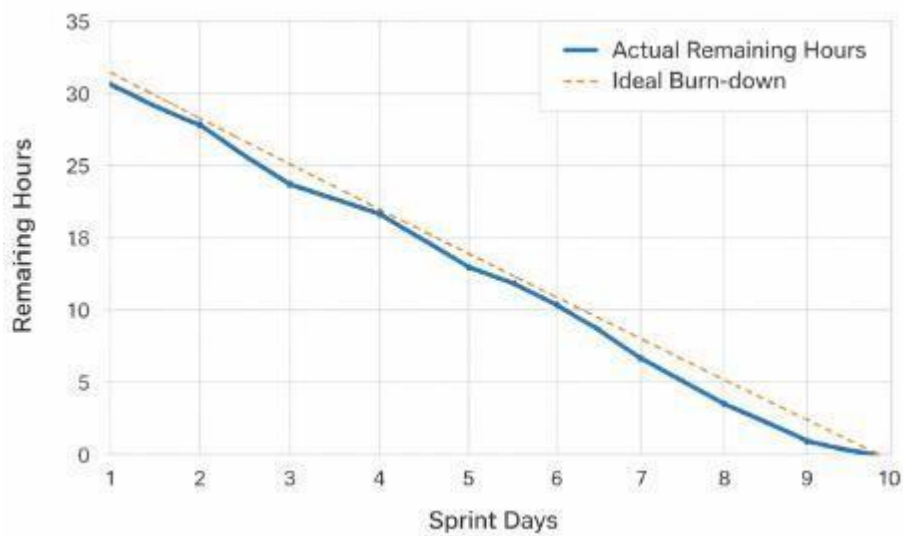
| Day | Remaining Hours |
|-----|-----------------|
| 1   | 32              |
| 2   | 28              |
| 3   | 25              |
| 4   | 20              |
| 5   | 16              |
| 6   | 12              |

|    |   |
|----|---|
| 7  | 9 |
| 8  | 5 |
| 9  | 2 |
| 10 | 0 |

4. Insert a **line chart**:

- X-axis = Sprint Days
- Y-axis = Remaining Hours
- Plot actual progress against ideal trend line

Optional: Use Trello Power-Ups like Corrello for automated burn-down charts.



## EXERCISE

**Project:** Simple Calculator Web Application

**Instructions:**

1. Define **Sprint Goal**
2. Prepare **Product Backlog**: Addition, Subtraction, Multiplication, Division, Input Validation, Error Handling, Responsive UI
3. Break each story into tasks, estimate effort, assign team members
4. Create **Trello Sprint Board**
5. Draw **priority chart** and **burn-down chart**

**Deliverables:**

- Sprint Goal document
- Sprint Backlog table with tasks, assignments, estimated hours
- Trello board screenshot
- Priority and burn-down charts
- Reflection paragraph